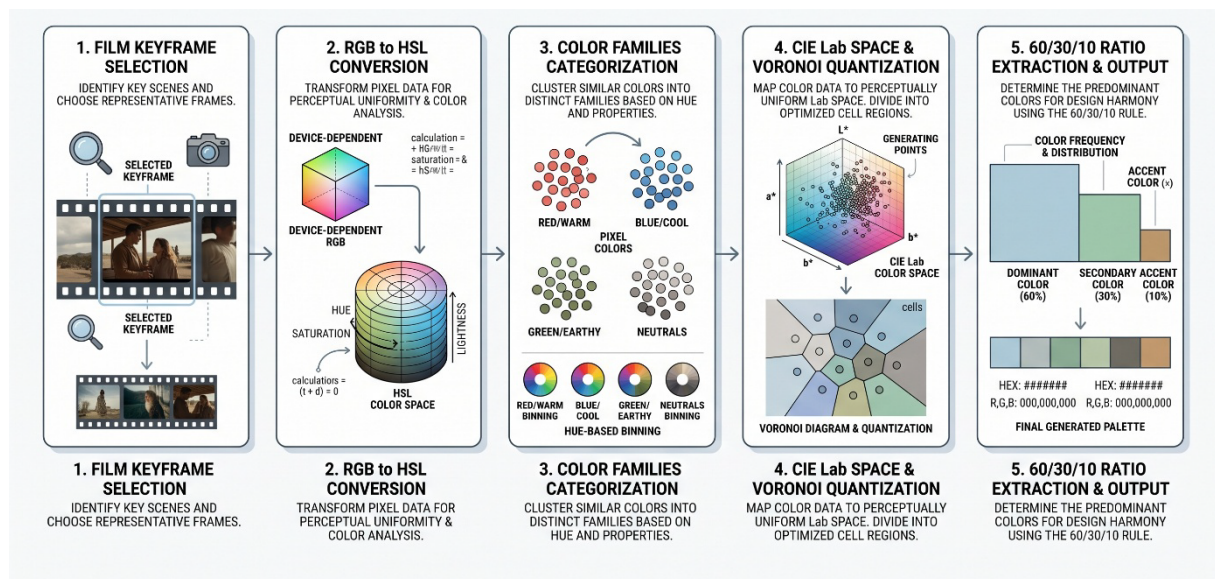


HSL and CIELAB Based Hybrid Perceptual Color Palette Analysis



In this study, a composite color analysis method called HSL and CIELAB Based Hybrid Perceptual Color Palette Analysis was used. The method evaluates colors not only according to numerical RGB values; It evaluates them together in terms of color family, saturation, luminance and perceptual color difference. Hybrid Perceptual Palette is not a singular and standard algorithm defined by the same name in the literature. Instead, it is a palette inference workflow defined within the scope of this study, which combines HSL-based color family preclassification, perceptual distance calculation in CIELAB space, and field assignment based on the nearest representative.

The RGB image is first read through the HTML5 Canvas and reduced to maintain speed of analysis. When activated by the user, the Dominant-Pixels based Mean Filter (DMF) is applied to reduce local pixel noise. This process limits JPEG artifacts, individual bright pixels, and small color fluctuations from unnecessarily splitting the palette. The non-transparent pixels are then analyzed according to the user-specified sampling density.

In the first stage, RGB pixels are converted to HSL space and divided into red, orange, yellow, green, teal, blue, purple, black, and white/gray color families. In color families, pixels with high saturation and close to medium luminance are evaluated with higher weight. Thus, instead of pale or extremely dark tones, the more distinctive colors of the scene stand out in representative color selection. In black and white/gray families, all of the relevant pixels are used.

The representative colors obtained for each family are converted into the CIE 1976 space. CIELAB allows for a more perceptually significant evaluation of the difference between colors relative to RGB channel values (Connolly & Fleiss, 1997). The family with the most pixels is determined as the dominant color; the secondary and accent colors are selected from candidates that are sufficiently separated from the selected centers. The selected representatives are used as fixed decision centers in CIELAB space, and

each pixel is assigned to the center that gives its lowest value. This process can be interpreted as Voronoi-type nearest-center zoning and is related to the quantization approach (Lloyd, 1982). $L^*a^*b^*\Delta E_{ab}^*$

CIELAB K-Means fallback is applied to prevent black, gray, and white tones from merging into the same color family in achromatic frames. Finally, clusters are ranked as dominant, secondary, and accent colors based on their pixel fill rate; for the tricolor palette, the results are compared to the 60 percent, 30 percent, and 10 percent targets. The CCT value, on the other hand, is not a clustering criterion, but an auxiliary indicator that supports the temperature character of the colors and is calculated using McCamy's (1992) approach.

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